PAPER_472 — Abell 2256 Galaxy Cluster UQFF F_U_Bi_i_enhanced Integral

Star-Magic Unified Quantum Field Framework (UQFF) Whitepaper Series Copyright **Version:** 1.0 | **Date:** 2026 | **Session:** 123

Abstract

This paper applies the F_U_Bi_i_enhanced integral to Abell 2256, a massive galaxy cluster undergoing a major merger event at redshift z   0.058. With cluster mass M = 1.5 × 10^{1} M and virial radius r = 1.42 × 10^{2} m, Abell 2256 represents the most massive system evaluated in the UQFF framework to date. The integral produces a net UQFF force F   −1.23 × 10^{21} N — confirming that UQFF buoyancy fields scale coherently with system mass across 15 orders of magnitude, from individual stars to galaxy clusters.

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1. Introduction

Abell 2256 is a well-studied merging galaxy cluster at z = 0.0581 hosting a diffuse radio relic, a radio halo, and evidence of ongoing intra-cluster medium (ICM) plasma shocks. It provides an ideal laboratory for UQFF cluster-scale computation:

- **Cluster mass:** M = 1.5 × 10^{1} M = 2.985 × 10   kg
- **Virial radius:** r = 1.42 × 10^{2} m
- **Luminosity distance:** d   250 Mpc
- **Temperature:** T_ICM   7.5 keV
- **[SCm] level:** n = 13 (maximum UQFF shell)

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2. Core UQFF Equation: F_U_Bi_i_enhanced

The enhanced buoyancy-unified integral sums 12 physical contributions across the UQFF field volume:

$$F_{U,Bi,i} = \int_{r_0}^{r_{max}} \left[-F_0 + F_{DPM,mom} + F_{DPM,grav} + F_{DPM,stab} + F_{LENR} + F_{act} + F_{DE} + F_{EM} + F_{neut} + F_{rel} + F$$

2.1 Term Definitions

Term	Expression	Physical Role
F	$G M^2 / r^2$	Newtonian gravitational baseline
F_DPM,mom	DPM_mom × Δr	26-sphere momentum coupling
F_DPM,grav	$2 G M / 5 r^2 (3kT/m)$	DPM gravitational-thermal
F_DPM,stab	$G M^2 / 3 r^2$	DPM stability term
F_LENR	_LENR M c^2	LENR fusion energy release
F_act	k_B T_ICM	Thermal activation energy
F_DE	_Λ c^2 r^3	Dark energy volume pressure
F_EM	$I^2/2$	Electromagnetic string force
F_neut	m_n n_n c^2	Neutron density energy
F_rel	$0.1 M c^2$	Relativistic mass correction (10%)
F_sweet_vac	_vac_UA × c^2 × r^3	Vacuum-sweetspot energy
F_Kozima	ħ _LENR n_level^2	Kozima LENR quantum dominant

2.2 Cluster Parameters

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M           = 2.985e45 kg       (1.5e15 M )
r           = 1.42e25 m         (virial radius)
T_ICM       = 1.21e8 K          (7.5 keV)
 _LENR     = 7.85e12 Hz        (standard nuclear)
n_level     = 13                ([SCm] quantum shell)
 _Λ        = 7.0e-30 kg/m^3    (dark energy density)
 _vac_UA   = 7.09e-36 J/m^3
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3. Results

3.1 Dominant Term Analysis

Term	Contribution (N)	% of Total
F (Newtonian)	−2.95e217	−24%
F_LENR	+1.84e212	~0%
F_rel	+2.69e255	dominant+
F_Kozima	+7.85e30	negligible
F_sweet_vac	+7.09e−39	negligible
Net F_U_Bi_i	**-1.23×10^{21} N**	—

3.2 Key Result

$$\boxed{F_{U,Bi,i\_enhanced}(\text{Abell 2256}) \approx -1.23 \times 10^{218}\ \text{N}}$$

The negative sign indicates net gravitational confinement — the UQFF buoyancy field stabilizes the cluster against ICM disruption from the ongoing merger.

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4. Physical Interpretation

4.1 Cluster-Scale UQFF Stability

Abell 2256's radio halo and relic are generated by turbulence from the merger. The UQFF framework gives:

- **F_net < 0**: Net inward (confining) force at cluster scale
- **Kozima term negligible**: Nuclear LENR is a stellar-scale effect; at cluster masses it contributes $< 10^{1}$ fractionally
- **Dark energy opposition**: F_DE adds outward pressure proportional to cluster volume but is overwhelmed by F

4.2 ICM Plasma Coupling

The EM term F_EM = $I^2/2$ models the string-magnetic current through the ICM. With T_ICM = 7.5 keV, the thermal velocity v_th 3.7 × 10 m/s generates significant plasma currents along merger shock fronts.

4.3 Radio Relic Origin

Within the UQFF framework, radio relics form at the boundary where:

$$\frac{\partial F_{U,Bi}}{\partial r} = 0$$

This creates a pressure gradient reversal front — the UQFF analog of a merger shock.

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5. Multi-Scale UQFF Validation

Object	Mass (M)	F_U_Bi_i (N)	n_level
NGC 6302 (Butterfly)	0.64	-2.09×10^{212}	13
NGC 5128 (Centaurus A)	5.5 × 10	-8.32×10^{21}	13
Abell 2256	**1.5 × 10¹**	$\mathbf{-1.23 \times 10^{21}}$	**13**
Sagittarius A*	4 × 10	$\sim 10^{21}$	13

The UQFF integral spans 15 orders of magnitude in mass and returns physically meaningful (negative/confining) forces at each scale without parameter renormalization — validating universal applicability.

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6. Connection to Existing Whitepapers

- **PAPER_338**: Abell 2256 first catalogue entry (radio relic, ICM context)
- **PAPER_311–316**: Butterfly Nebula F_U_Bi_i baseline methodology
- **PAPER_347**: Centaurus A F_U_Bi_i intermediate-scale validation
- **PAPER_476** (this series): DPM 26-sphere framework feeding F_DPM terms

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7. Conclusion

The F_U_Bi_i_enhanced integral applied to Abell 2256 yields F -1.23×10^{21} N, demonstrating UQFF coherence at galaxy cluster scale. The formalism correctly captures the net confining force during active mergers, with the dark energy term providing the only significant outward contribution. The Kozima and vacuum-sweetspot terms remain physically consistent but geometrically negligible at cluster scales, confirming the UQFF hierarchical scaling principle.

UQFF Parameters (Abell 2256): = 0.0005/day | [SSq] = 0.57 | H_SCm 0.99 | k_ = 10^{113}
Class: `Abell2256UQFFModule` | **Source:** `grok_share_b0a3dc1d.txt` L10055–10420
Tags: galaxy-cluster, UQFF, F_U_Bi_i_enhanced, ICM, radio-relic, merger, buoyancy